Prepared By	09-Jun-2012	Mr.Chatchawan S.	RNC RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Checked By	09-Jun-2012	Mr.Chatchawan S.	
Approved By	09-Jun-2012	Mr.Amnauy W.	
Certified By	09-Jun-2012	Mr.Amnauy W.	Renew coils Engineering & Supply Co., Ltd.

Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-009-2012

Addendum

WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WS-002

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name	**Renew Coil Engineering & Supply Co., Ltd.**		
WPS No.	RNC-WS-002	Date 23-Feb-13	Revision No. 1
Supporting PQR No.(s)	RNC-PQR-002	Date 23-Feb-13	Type(s) Manual
Welding Process(es)	Gas Tungsten Arc Weleing (GTAW)		(Automatic, **Manual**, Machine, or Semi-Auto)

JOINT (QW-402)

Joint Design: **Single V**

Backing (Yes) ____ (No) X

Backing Material (Type): N/A

☐ Metal ☐ Nonfusing Metal ☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. 1 Group No. 1 or 2 to P-No. 1 Group No. 1 or 2

Material Specification Type and Grade: A106 Gr.B, A234 WPB Equivalents TO A106 Gr.B, A234 WPB Equivalents

Chem. Analysis and Mech. Prop.: -

Thickness Range :

Base Metal :	Groove	1.6 mm - 17.12 mm.	Fillet	All
Pipe Dia. Range :	Groove	All	Fillet	All

Other

FILLER METALS (QW-404)

	GTAW	
Spec. No. (SFA)	5.18	
AWS No. (Class)	ER 70S-G	
F-No.	6	
A-No.	1	
Size of Filler Metals	Ø 2.0 & 2.4 mm	
Deposited Weld Metals	8.56 mm.	
Thickness Range :		
Groove	17.12 mm Max	
Fillet	All	
Electrode-Flux (Class)	-	
Flux Trade Name	-	
Consumable Insert	-	
Other	Kobelco TGS-50 or Equivalents	

POSITION (QW-405)

Position(s) of Groove: ALL

Welding Progression: Up X Down

Position(s) of Fillet: ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: N/A

Time Range: N/A

Notes: T>20 mm heat treatment 593-649 Holding 1 hr

PREHEAT (QW-406)

Preheat Temp. Min. T < 25 mm. 10° C

Interpass Temp. Max. T ≥ 25 mm. - 79° C

Preheat Maintenance: N/A

Notes

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	10-20 Ltr/Min
Trailing	-	-	-
Backing	-	-	-

Sheet 1 of 2

RNC **WELDING PROCEDURE SPECIFICATION (WPS)**

WPS No. RNC-WS-002 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current [ ] AC [X] DC Polarity EN

Amps (Range) 105-119 Volts (Range) 9-12

Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead STRING OR WEAVE

Orifice or Gas Cup Size NO. 4-9

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE

Travel Speed (Range) 4.97-9.23 Cm/Minute

Peening N/A

Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	100-105	9-10	4-11	
2nd	GTAW	ER 70S-G	2.4	DC EN	105-108	9-10	4-11	
Other(s)	GTAW	ER 70S-G	2.4	DC EN	105-119	10-12	4-11	

We,the undersigned, certify that the statements in this record are correct and that the test welds were prepared ,welded, and tested in accordance with the requirement of **ASME-IX, 2010 latest Revision.**

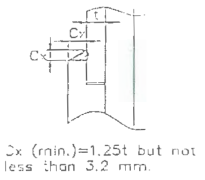
Prepared by : Mr. Chatchawan. S (Name) [signature] (Signed) 23/02/13 (Date)

Reviewed by : ภูชิน สีหอมไชย Phuchin Seehomchai (Name) [signature] (Signed) 11/03/13 (Date)

Approved by: ภูชิน สีหอมไชย Phuchin Seehomchai (Name) [signature] (Signed) 11/03/13. (Date)

Sheet 2 of 2

ATTACHED FIGURE JOINT SKETCH



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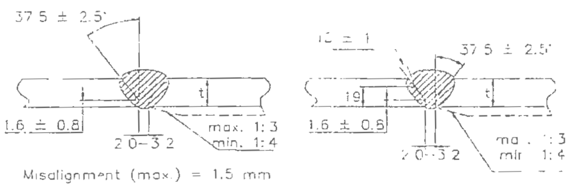

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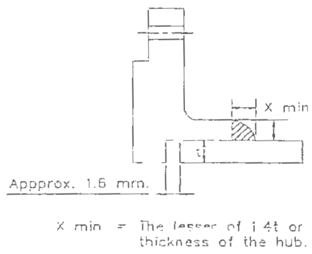

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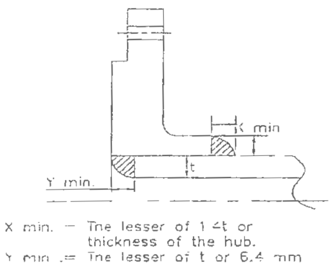

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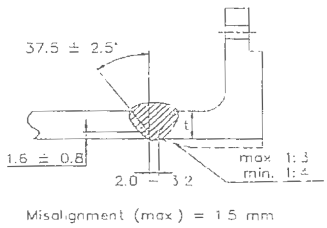

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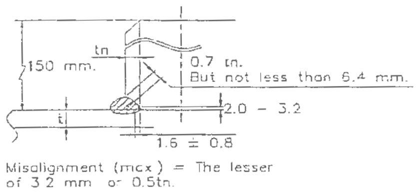

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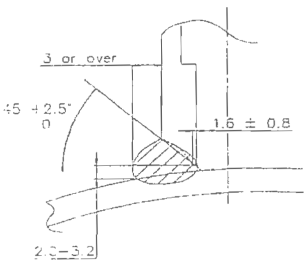

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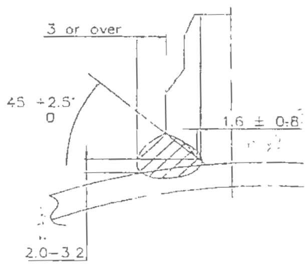

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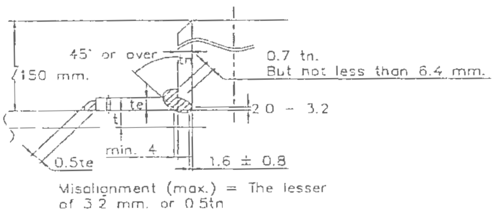

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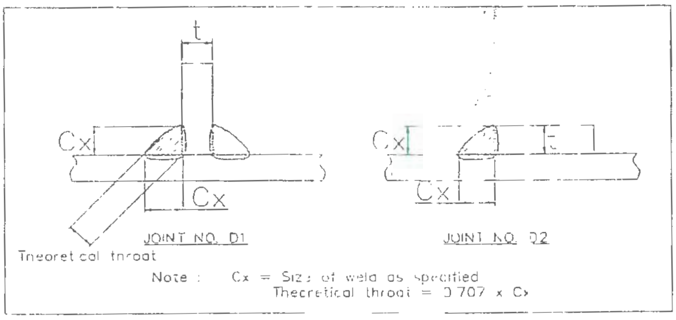


Procedure Qualification Record (PQR)

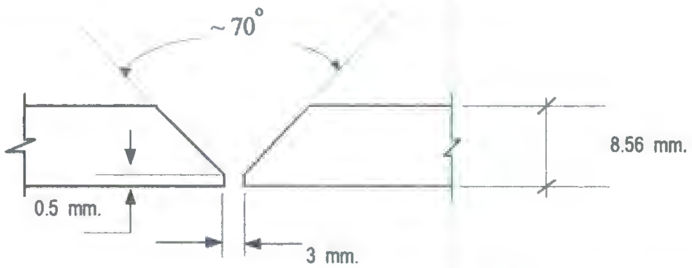
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-002

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-002	Date :	23 February 2013
WPS Number :	RNC-WS-002		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic):	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)

Material Spec.	A 106	To	A 106
Type or Grade	Gr.B	To	Gr.B
P. No.	1	To	1
Thickness of Test Coupon	8.56 mm.		
Diameter of Test Coupon	Ø 4"		
Others			

FILLER METALS (QW-404)

SFA Specification	A 5.18
AWS Classification	ER70S-G
Filler Metal F-No.	6
Weld Metal Analysis A-No.	1
Size of Filler Metal	Ø 2.4 mm.
Others	Kobelco TGS-50
Weld Metal Thickness	-

POSITION (QW-405)

Position of Groove	6G
Progression of Welding (Uphill or Down)	Up Hill
Others	N/A

PREHEAT (QW-406)

Preheat Temperature	33° C
Interpass Temperature (max.)	179° C
Others	N/A

POSTWELD HEAT TREATMENT (QW-407)

Temperature	N/A
Hold Time	N/A
Others	

GAS (QW-408)

	Percent Composition		
	Gas (es)	Mixture	Flow Rate
Shielding	Argon	99.99%	11 L/Min.
Trailing	-	-	-
Backing	-	-	-

ELECTRICAL CHARACTERISTICS (QW-409)

Current	DC		
Polarity	EN		
Amperage	105 - 119	Voltage	10 - 10
Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2		
Others	N/A		

TECHNIQUE (QW-410)

Travel Speed	4.97 - 9.28 cm/min.
String or Weave Bead	Both
Oscillation	N/A
Multiple or single Pass	Multiple Pass
Multiple or Single Electrodes	Single Electrode
Others	N/A

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number-: RNC-PQR-002

Tensile Test Report No.TS-13-02-083

Specimen	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (KN)	Ultimate Strength (N/mm^2)	Type of Failure & Loc.
544/13 TR1	19.13	8.14	155.72	85.10	546.51	OUT OF WELD
544/13 TR2	19.11	8.28	158.230	85.80	542.22	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-032

Type and Figure Number	Type of Bend	Indication	Results
544/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
544/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
544/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
544/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-006

Hardness Testing Report No. HV-13-02-014

Macrostructure Report No. MA-13-02-048

Impact Testing Report No.IM-13-02-091

Welders Name Mr.Prathom M. Welder Number W-009

Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date : 24/02/13

Inspected by : Mr.Wichit K.

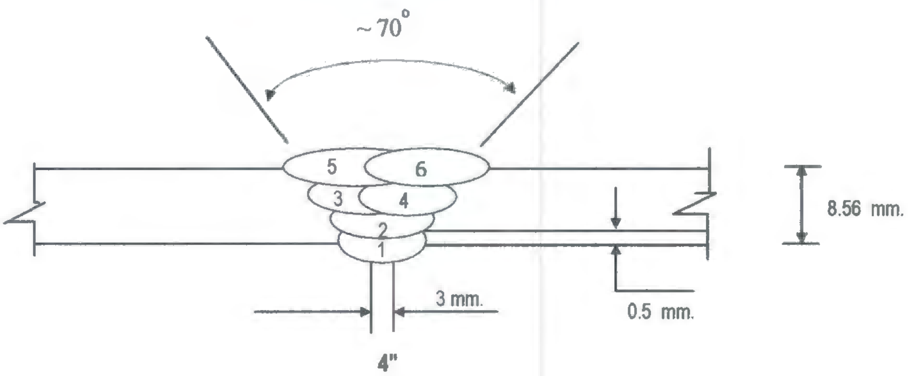
Certified by : Mr.Chalermkiat R.

Date : 23 February 2013

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Welding Sequences for PQR-RNC-002 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	105	10	DCEN	-	-	4.97	12.68	11	-
2	GTAW	ER70S-G	2.4	108	10	DCEN	-	-	9.28	6.99	11	-
3	GTAW	ER70S-G	2.4	106	10	DCEN	-	-	7.62	8.35	11	-
4	GTAW	ER70S-G	2.4	119	10	DCEN	-	-	7.36	9.71	11	-
5	GTAW	ER70S-G	2.4	118	10	DCEN	-	-	6.24	11.35	11	-
6	GTAW	ER70S-G	2.4	116	10	DCEN	-	-	6.10	11.41	11	-

PAE Technical Service Company Limited

Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

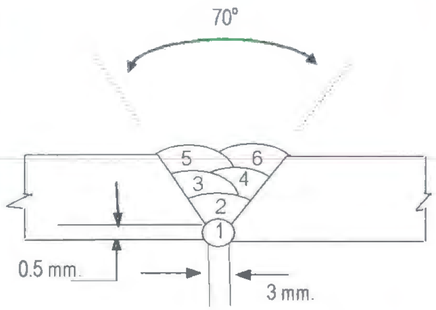
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Client :	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-018
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-002	Date : 13-Feb-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-009	MR.PRATHOM M.	GTAW	6G	ASME IX	A106 Gr.B	ø 4" Thk. 8.56 mm.	A5.18	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	105	10	4.97	33	12.68	Argon	11	-	-
2	GTAW	ER70S-G	2.4	DCEN	108	10	9.28	91	6.99	Argon	11	-	-
3	GTAW	ER70S-G	2.4	DCEN	106	10	7.62	103	8.35	Argon	11	-	-
4	GTAW	ER70S-G	2.4	DCEN	119	10	7.36	179	9.71	Argon	11	-	-
5	GTAW	ER70S-G	2.4	DCEN	118	10	6.24	150	11.35	Argon	11	-	-
6	GTAW	ER70S-G	2.4	DCEN	116	10	6.10	153	11.41	Argon	11	-	-

PAE Inspection By		Witness By		Approved By	
SIGNATURE :		SIGNATURE :		SIGNATURE :	
NAME :	Mr.Wichit K.	NAME :	Mr. Chatchawan	NAME :	ภูชิน สีหอมไชย Phuchin Seehomchai
DATE :	14-Feb-2013	DATE :	19/02/13	DATE :	

PAE TECHNICAL SERVICE

RENEW COILS ENGINEERING & SUPPLY QA/QC RNC

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakk ongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax 0 2815 6415 , E-mail : qa@lpnpm.co.th

29779

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1

Report No. TS-13-02-083

Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 4" (8.56 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment / Serial No.	INSTRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm^2)	Gauge Length (mm)	Load (kN)	Strength (N/mm^2)	Load (kN)	Strength (N/mm^2)		
1	544/13 TR1	N/A	8.14	19.13	155.72	N/A	N/A	N/A	85.10	546.51	N/A	OUT OF WELD
2	544/13 TR2	N/A	8.28	19.11	158.23	N/A	N/A	N/A	85.80	542.22	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. RNC QA/QC RNC

Reported by : N. [signature]

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : N. [signature]

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

29785

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-032

Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	∅ 4" (8.56 mm Thk.)	Diameter of Meudrel	32 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	544/13 BF1	Transverse face bend	8.14x19.14x160.73 mm	No Open Discontinuity	N/A	-
2	544/13 BF2	Transverse face bend	8.11x19.15x160.77 mm	No Open Discontinuity	N/A	-
3	544/13 BR1	Transverse root bend	8.14x19.17x160.46 mm	No Open Discontinuity	N/A	-
4	544/13 BR2	Transverse root bend	8.68x19.14x160.55 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

0791

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-02-091

Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a[ε1]
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	∅ 4" (8.56 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	19 February 2013	Equipment 's Capacity	542 Joules
Tested Date	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension Thickness (mm)	Width (mm)	Length (mm)	Results Energy Absorbed (Joules)	Average (Joules)	Location of Test	Remark
1	544/13-1	-46°C	6.713	10.006	54.88	14.68	26.33	Weld Metal	-
2	544/13-2	-46°C	6.713	10.004	54.87	51.75			
3	544/13-3	-46°C	6.713	10.008	54.84	12.57			
4	544/13-1	-46°C	6.713	10.007	54.91	80.46	92.08	Base Metal	-
5	544/13-2	-46°C	6.714	10.008	54.86	100.51			
6	544/13-3	-46°C	6.713	10.010	54.90	95.27			
7	544/13-1	-46°C	6.713	10.008	54.95	175.12	167.41	HAZ	-
8	544/13-2	-46°C	6.714	10.009	54.98	179.75			
9	544/13-3	-46°C	6.714	10.008	54.96	147.36			

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. RNC QA/QC RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.

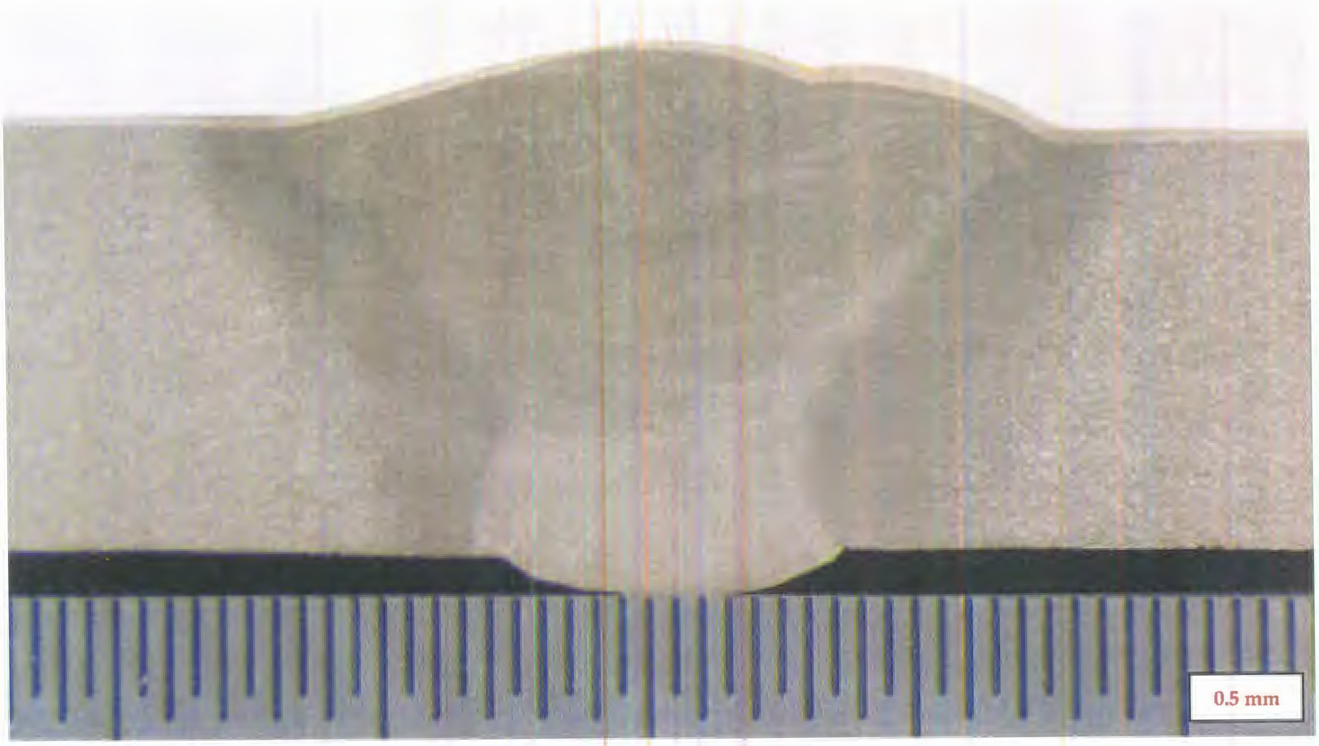
Testing Laboratory LPN Plate Mill Public Company Limited
199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.
Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th 3695

Macrostructure Analysis Report

Report No. MA-13-02-048
Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Test No.	544/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A106 Gr.B	Method of Preparation	ASME Secction IX : 2010 edition
Dimension	4" (8.56 mm Thk.)	Test Method	ASME Secction IX : 2010 edition
Welding Process/Position	GTAW/6G	Standard of Test	ASME Secction IX : 2010 edition
Type of Joint	Butt Jiont	Etchant Reagent	25 ml HNO_3 + 75 ml H_2O
Welder Name / No.	Mr. Prathom M. / W-009	Time of Etching	5 minutes
Date Received	18 February 2013	Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

QA/QC RNC
RNC RENEW COILS ENGINEERING CO.,LTD.

Reported by : N.
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13

Approved by : N.
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 22/02/13

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited
199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.
Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

13231

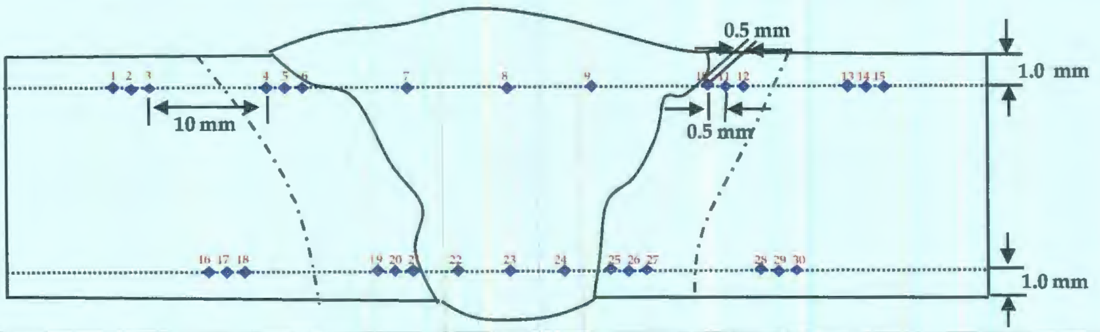
Hardness Testing Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Report No. HV-13-02-014
Revision No. 0

Test Product	WPS No. RNC-WPS-002	Material Characterization	Pipe
Test No.	544/13 HN	Test Method	ASTM E92-82 (Reapproved 2003)$^{\varepsilon 2}$
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Secction IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A106 Gr.B	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	4" (8.56 mm Thk.)	Indentor Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Jiont	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0 °C

Test Location
Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	147.5	16	BASE	154.8		***************************	
2	BASE	148.3	17	BASE	152.5			
3	BASE	148.5	18	BASE	151.8			
4	HAZ	166.7	19	HAZ	162.2			
5	HAZ	176.4	20	HAZ	165.6			
6	HAZ	189.2	21	HAZ	169.2			
7	WELD	196.6	22	WELD	155.4			
8	WELD	191.2	23	WELD	154.6			
9	WELD	192.2	24	WELD	153.9			
10	HAZ	188.4	25	HAZ	164.6			
11	HAZ	177.4	26	HAZ	166.9			
12	HAZ	175.0	27	HAZ	162.0			
13	BASE	151.8	28	BASE	152.4			
14	BASE	152.4	29	BASE	152.5			
15	BASE	152.6	30	BASE	149.8			

Additional details : -

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Reported by : N. ...
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13

Approved by : N. ...
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 22/02/13

LPN PLATE MILL

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ArcelorMittal South Africa Limited
Tubular Products
Genl. Hertzog rd.
P O Box 48 Vereeniging 1930
South Africa

MATERIAL TEST CERTIFICATE
SEAMLESS TUBE

Telephone +27 (0)16 450 4220
Fax +27 (0)16 423 4906

Licensed under API Spec 5L and 5CT
5L - 0319
5CT - 0419

ISO 9001: 2008
DE-395997 QM
Test Certificate EN 10204:2004 TYPE 3.1

ArcelorMittal

Customer:	UR MENA LIMITED	Customer Order/Contract No:	STH7 136520
Order No:	4000012782	Material No:	1000000403
Certificate Reference No:	040060162150	Cast/Heat No:	98B2332
Product:	FULLY KILLED HOT FINISHED CARBON STEEL SEAMLESS TUBES		Page 1 of 1
Specification:	ISO3183:2007/API 5L:2007 L245/L290/B/X42 PSL1 ASTM A106.08B/A53B.07/ A530.04 ASME SA106.08B/SA53B.07/SA530.04		
Product Marking:	ARCELORMITTAL SA ISO3183/Spec 5L-0319 MONOGRAM 07-11 ASTM/ASME A/SA106B A/SA53B 114.300 (4.500) 8.560 (0.337) 6.000 (19.700) L245/B L290/X42 PSL1 SMLS TESTED 20.7 MPa (3000 psi) CAST NO:98B2332 PROD/O NO: T7741278210		

General Information

Quantity	Mass	Dimensions			Total Length
		Tube OD	Thickness	Length	
59	7,901.280(kg)	114.300(mm)	8.560(mm)	6.000(m)	354.000(m)
	17,419.162(lb)	4.500(")	0.337(")	19.700 (ft)	1,162.300(ft)

Steel making process	Final Rolling Operation
Electric Arc	Final hot rolling operation finished above 860°C and cooled in still air.

Chemical Composition

R22-(V+Nb+Ti) R24-(Nb+V)

Element(%)	C	Si	Mn	S	P	Cr	Ni	Mo	Cu	V	Al	Ti	Sn	Ca	N	B	Nb	CE	R22	R24
Minimum	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Maximum	0.200	-	1.30	0.030	0.030	0.50	0.50	0.150	0.500	-	-	-	-	-	-	-	-	0.41	0.15	0.060
Heat	0.171	0.23	0.86	0.003	0.014	0.12	0.09	0.038	0.110	0.005	0.043	0.001	0.006	0.0012	0.0089	0.0002	0.0020	0.36	0.01	0.007
Product	0.171	0.230	0.860	0.003	0.014	0.120	0.090	0.038	0.110	0.005	0.043	0.001	0.006	0.001	0.009	0.000	0.002	0.360	0.008	0.007
Product (ADD)	0.171	0.230	0.860	0.003	0.014	0.120	0.090	0.038	0.110	0.005	0.043	0.001	0.006	0.001	0.009	0.000	0.002	0.360	0.008	0.007

PTT Global Chemical
Chayut S.
Inspection QA/QC Division
Integrity and Reliability
27/2/2013

Mechanical Properties

Specification Limits	UTS (Rm) MPa	UTS (Rm) psi	Yield (0.5%) MPa	Yield (0.5%) psi	% EL 50mm
Minimum	415	60000	290	42000	30.0
Maximum	-	-	-	-	-
(1) Actual	506	73389	369	53519	36.0
(2) Actual	502	72808	365	52938	37.0
(3) Actual					
(4) Actual					

	UTS (Rm) MPa	UTS (Rm) psi	Yield (0.5%) MPa	Yield (0.5%) psi	% EL 50mm
(5) Actual					
(6) Actual					
(7) Actual					
Orientation & type of tensile test	Longitudinal, Strip				
Width of tensile piece (mm)	25.4mm				
Orientation of impact test piece					

QA/QC RNC
RNC Revco Engineering Supply Co.,Ltd.

OTHER TESTS	
Flattening	Passed
Hydrostatic	20700 kPa (3000 psi) - 5 sec
NDI: EMI	PASS - ASTM E570 - 12.5% NOTCH
NDI: UT	UT not required
HV 10 Kg	150 154 156

COPY

PURCHASE ORDER NO. 0045017909
ITEM NO. 4

UR MENA LIMITED Q.A. TEL: 0-2138-8200

Remarks:

Material in accordance with NACE MR0175:2003/ISO15156-2:2003, MR0103:2010. Dimensions to ANSI B36.10M-2004
The material conform to the hot yield strength requirements as per ASME, Sect II, Pt D, Table Y-1, 2010
All the material conform to the visual and dimensional requirements and is made to a suitable fine grain practice

Quality Assurance Manager: R. Bester *Date of Issue: 2011.07.29* *Certified by:* RB

We hereby certify that the steel grade and quality level of all products are in conformity with the order and comply fully with specification requirements. No changes, amendments or additions may be made to this document. Any changes which are effected shall invalidate this certificate.

KOBELCO

INSPECTION CERTIFICATE

Date of Issue	17/Feb/2011
Certificate No.	KF11-002
Date of Inspection	15/Feb/2011

...RCHASER

...RADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
TG-S50	2.4	F0B2704	AWS A5.18 ER70S-G

Chemical Composition (%)

Elements	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V
WIRE	0.10	0.75	1.39	0.011	0.017	0.13	0.01	0.01	< 0.01	0.002

Tensile Test of Weld Metal			Impact Test of Weld Metal			
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy		Tested size for Mechanical test: 2.4 mm Tested MFG. No. of
453 MPa	545 MPa	38 %	-30 °C	Avg.	257 J	Mechanical test: F9B4140

Welding Conditions			
...Current	DCEN	Shielding Gas	Ar
...ding Current	128 A	-------	-------
...ding Voltage	14 V	-------	-------

REMARKS

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD.
QA/QC
RNC

...hereby certify that the test results of the above welding ...aterial are as described herein and satisfy the requirements ...applicable specification.

KOBE MIG WIRE (THAILAND) CO., LTD.

CHIEF INSPECTOR

RNC

RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name: **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. RNC-WS-002 Date 10-Oct-12 Revision No. 1

Supporting PQR No.(s) RNC-PQ-002 Date 25-May-09 Type(s) Manual

Welding Process(es) Gas Tungsten Arc Welding (GTAW) (Automatic, **Manual**, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design: **Single V**

Backing (Yes) ______ (No) X

Backing Material (Type): N/A

☐ Metal ☐ Nonfusing Metal ☐ Nonmetallic ☐ Other

Notes ______

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. 1 Group No. 1 or 2 to P-No. 1 Group No. 1 or 2

Material Specification Type and Grade: A106 Gr.B, A234 WPB Equivalents TO A106 Gr.B, A234 WPB Equivalents

Chem. Analysis and Mech. Prop. ______ -

Thickness Range :

Base Metal : Groove 2.0 mm - 14.22 mm. Fillet All

Pipe Dia. Range : Groove All Fillet All

Other ______

FILLER METALS (QW-404)

FILLER METALS (QW-404)	GTAW	
Spec. No. (SFA)	5.18	
AWS No. (Class)	ER 70S-G	
F-No.	6	
A-No.	1	
Size of Filler Metals	Ø 2.0 & 2.4 mm	
Deposited Weld Metals	7.11 mm	
Thickness Range :		
Groove	14.22 mm Max.	
Fillet	All	
Electrode-Flux (Class)	-	
Flux Trade Name	-	
Consumable Insert	-	
Other	Kobelco TGS-50 or Equivalents	

POSITION (QW-405)

Position(s) of Groove: ALL

Welding Progression: Up X Down ______

Position(s) of Fillet: ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: N/A

Time Range: N/A

Notes ______

PREHEAT (QW-406)

Preheat Temp. Min. 10°C

Interpass Temp. Max. 250°C

Preheat Maintenance: N/A

Notes ______

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	10-20 Ltr/Min
Trailing	-	-	-
Backing	-	-	-

Sheet 1 of 2

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WS-002 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current [ ] AC [X] DC Polarity EN

Amps (Range) 110-170 Volts (Range) 16-25

Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead STRING OR WEAVE

Orifice or Gas Cup Size NO. 4-9

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE

Travel Speed (Range) 4-11 Cm/Minute

Peening N/A

Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	110-150	16-22	4-11	
2nd	GTAW	ER 70S-G	2.4	DC EN	130-170	18-25	4-11	
Other(s)	GTAW	ER 70S-G	2.4	DC EN	120-170	16-25	4-11	

We,the undersigned, certify that the statements in this record are correct and that the test welds were prepared ,welded, and tested in accordance with the requirement of **ASME-IX, 2007 latest Revision.**

	Name	Signed	Date
Prepared by :	Mr. Chatchawan. S	[signature]	10/10/12
Reviewed by	Mr. Seksan j	[signature]	10-10-12
Approved by:	Amnuay W.	[signature]	10-10-12

RENEW COILS ENGINEER & SUPPLY CO.,LTD. QC/QA RNC

Sheet 2 of 2

KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Welding Procedure Specification (WPS)

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name: RENEW COILS ENGINEERING LIMITED PARTNERSHIP
WPS No. KWC-06A0017-WPS-RNC-PN1   Ref. PQR No. KWC-06A0017-PQR-RNC-PN1
Revision No. 0   Date. JULY 20, 2006
Welding Process(es) GTAW and GMAW   Type(s) MANUAL
(Automatic, Manual, Machine, or Semi-auto)

JOINTS (QW - 402)

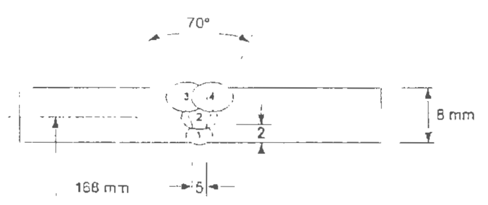
Joint Design: 6G
Backing (Yes) ____ (No) ✓
Backing Materials (Type) -
(Refer to both backing and retainers)

☐ Metal ☐ Nonfusing metal
☐ Non-metallic ☐ Other

Sketches, Production Drawings, Weld Symbols or Written Description should show the general arrangement of the parts to be welded. Where applicable, the root spacing and the details of weld groove may be specified.

(At the option of the Mfgr., sketches may be attached to illustrate join design, weld layers and bead sequence, e. g., for notch toughness procedures, for multiple process procedures, etc.)

DETAILS



*BASE METALS (QW - 403)

P No. 1   Group No. 1   To P No. 1   Group No. 1
OR
Specification type and grade: A106 Gr B
to Specification type and grade: A106 Gr B
OR
Chem. Analysis and Mech. Prop. -
to Chem. Analysis and Mech. Prop. -
Thickness Range:
Base Metal: Groove 3-16 mm   Fillet -
Other

*FILLER METALS (QW - 404)

Spec. No. (SFA)	5.18	5.1
AWS No. (Class)	ER 70S-6	E 7016
F-No	6	4
A-No	1	1
Size of Filler Metals	Ø2.4mm	Ø3.2mm
Weld Metal		
Thickness Range:	4 mm (max)	12 mm (max)
Groove	4 mm (max)	12 mm (max)
Fillet	All	All
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	-	-

* Each base metal-filler metal combination should be recorded individually.

(R. Phaonium)
Designed

(S. Peansukmanee)
Approved

Authorized
Manufacturer

This document is not official unless it carries the authorized signatures.

KMUTT'S WELDING RESEARCH AND CONSULTING CENTER

Procedure Qualification Records (PQR)

PROCEDURE QUALIFICATION RECORDS (PQR)

Company Name RENEW COILS ENGINEERING LIMITED PARTNERSHIP

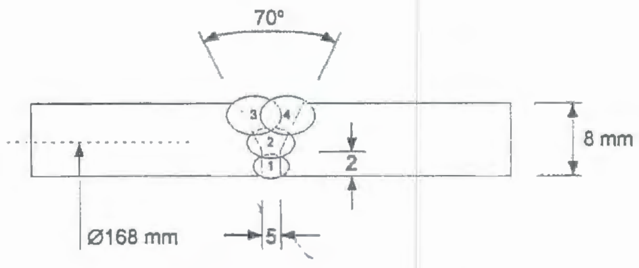
Procedure Qualification No. KWC-06A0017-PQR-RNC-PN1 Date JULY 20, 2006

Pre-Qualified WPS No. KWC-06A0017-WPS-RNC-PN1 Revision No. 0

Welding Process(es) GTAW+SMAW

Types (Manual, Automatic, Semi-Auto.) MANUAL

JOINTS (QW – 402)



Groove Design of Test Coupon

(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used.)

BASE METALS (QW – 403)

Material Spec. A106

Type or Grade Gr. B

P No. 1 to P No. 1

Thickness of Test Coupon 8 mm

Diameter of Test Coupon 168 mm

Other(s) PIPE

FILLER METALS (QW – 404)

SFA Specification 5.18 5.1

AWS Classification ER70S-6 E7016

Filler Metal F-No. 6 4

Weld Metal Analysis A-No. 1 1

Size of Filler Metal Ø2.4mm Ø3.2mm

Other -

Weld Metal Thickness 8 mm

POSITION (QW – 405)

Position of Groove 6G

Weld Progression (Up, Downhill) UPHILL

Other -

PREHEAT (QW – 406)

Preheat Temp. -

Interpass Temp. NOT EXCEED 100°C

Other -

POSTWELD HEAT TREATMENT(QW–407)

Temperature -

Time -

Other(s) -

GAS(QW–408) Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	Ar	-	20 lpm
Trailing	-	-	-
Backing	Ar	-	20 lpm

ELECTRICAL CHARACTERISTICS(QW–409)

Current DIRECT CURRENT

Polarity GTAW DC- SMAW DC-

Amp 80 100 Volt 13.2 32.2

Tungsten Electrode Size 2.4 mm

Other -

TECHNIQUE(QW–410) GTAW SMAW

Travel Speed 4-10 cm/min 5-15 cm/min

String or Weave Bead STRING WEAVE

Oscillation -

Multi or Single Pass (per side) MULTI PASS

Single or Multiple Electrode SINGLE

Other -

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KMUTT'S WELDING RESEARCH AND CONSULTING CENTER

Welding Procedure Specification (WPS) Cont'd

WPS (Back)

WPS No. KWC-06A0017-WPS-RNC-PN1 Rev. 0

POSITIONS (QW - 405)
Position(s) of Groove All
Welding Progression: Up ✓ Down
Position(s) of Fillet -

POST WELD HEAT TREATMENT (QW - 407)
Temperature Range -
Time Range -

PREHEAT (QW - 406)
Preheat Temp. Min. -
Interpass Temp. Max. 100°C
Preheat Maintenance -
(Continuous or special heating where applicable should be recorded)

GAS (QW - 408

	Percent Composition Gas(es)	Mixture(s)	Flow Rate
Shielding	Ar	-	20 lpm
Trailing	-	-	-
Backing	Ar	-	20 lpm

ELECTRICAL CHARACTERISTICS (QW - 409)
Current AC or DC DC
Polarity GTAW DC-, SMAW DC-
Amps (Range) GTAW 80, SMAW 100
Volts (Range) GTAW 13.2, SMAW 32.2

(Amps and volts range should be recorded for each electrode size, position, and thickness, etc. This information may be listed in a tabular form similar to that shown below.)

Tungsten Electrode Size and Type 2% Thotiated (EWTh-2)
(Pure Tungsten, 2% Thoriated, etc.)
Mode of Metal Transfer for GMAW -
(Spray arc, short circuiting arc, etc.)
Electrode Wire feed speed range -

TECHNIQUE (QW - 410)
String or Weave Bead GTAW STRING, SMAW WEAVE
Orifice or Gas Cup Size Ø10 mm
Initial and Interpass Cleaning (Brushing, Grinding, etc.) Brushing and Grinding
Method of Back Gouging -
Oscillation -
Contact Tube to Work Distance -
Multiple or Single Pass (per side) MULTIPLE (4)
Multiple or Single Electrodes SINGLE
Travel Speed (Range) GTAW 4-10 cm/min, SMAW 5-15 cm/min
Peening -
Other

Weld Layer(s)	Process	Filler Metal Class	Filler Metal Dia. (mm)	Current Type Polar.	Current Amp. Range (A)	Volt Range (V)	Travel Speed Range (cm/min)	Others (e.g., Remarks, Comments, Hot Wire Addition, Technique, Torch Angle, Etc.)
1	GTAW	ER70S6	2.4	DCEN	80-110	13.2-15 5	4-10	
2	SMAW	E 7016	3.2	DCEN	100-150	24-31	5-15	
3	SMAW	E 7016	3.2	DCEN	100-150	24-31	5-15	

(R. Phaonium)
Designed

(S. Peansukmanee)
Approved

Authorized
Manufacturer

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KMUTT'S WELDING RESEARCH AND CONSULTING CENTER

Procedure Qualification Records (PQR) Cont'd

PROCEDURE QUALIFICATION RECORDS (BACK)

TEST RESULTS OF TEST COUPON PQR No. KWC-06A0017-PQR-RNC-PN1

TENSILE TESTS (QW – 150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure & Location
PN1-TS01	22.00	4.90	107.80	69.550	645.189	Weld Area
PN1-TS02	22.90	5.60	128.24	72.350	564.176	Weld Area

GUIDED BEND TESTS (QW – 160)

Type & Figure No.	Result	Remarks
PN1-FB1, Face Bend	No Open Defect at Weldment	-
PN1-FB2, Face Bend	Open Discontinuity <3.2 mm at Weld	-
PN1-RB1, Root Bend	No Open Defect at Weldment	-
PN1-RB2, Root Bend	Open Discontinuity <3.2 mm at Weld	-

TOUGHNESS TESTS (QW – 170)

Specimen No.	Notch Location	Specimen Size	Test Temp.	Impact Values Ft. Lbs.	Impact Values Shear	Impact Values Mils	Drop Weight Break (Y/N)

Comments: ______

FILLET WELD TESTS (QW – 180)

Result-Satisfactory: YES____ NO____ | Penetration into Parent Metal: YES____ NO____

Macro Result: ______

OTHER TEST

Type of Test ______

Deposit Analysis ______

Other ______

Welder's Name ______ Clock No.____ Stamp No.____

Test Conducted by KINGWELD Laboratory Test No. KWC-06A0017

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Code.

Ekkachai Warinsiriruk

(E. Warinsiriruk)
Inspector

Authorized
Manufacturer

This document is not official unless it carries the authorized signatures.

RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

59 M.8 HIGHWAY NO. 36 T. TUBMA A. MUANG RAYONG 21000

Tel. : (038) 663476 FAX: (038) 663477 Mobile : (081) 8613795

WELDING PROCEDURE SPECIFICATIONS (WPS)

Company Name	RENEWCOILS ENGINEERING &SUPPLY CO.,LTD.	By	MR. VEERACHAT F. (Sign/Date)
Welding Procedure Specification no.	RNC-WS-002	Date	MAY 25, 2009
Revision no.	- Date -	Supporting PQR no.(s)	RNC-PQ-002
Welding Process(es)	1/ GTAW 2/ -	Type(s)	1/ Manual
Impact Test Requirement	☐ Yes ☑ No		2/ -

JOINTS (QW-402)

Joint Design: Butt joint (Single V-groove)

Backing: ☐ Yes ☑ No

Backing Material (Type) 1/ NA

2/ -

☐ Metal ☐ Nonfusing Metal

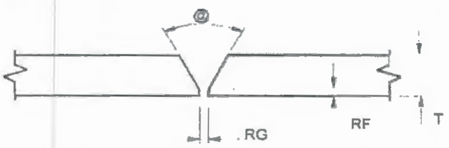
☐ Nonmetallic ☐ Other (Weld Metal)

Applicable of Tack Welding ☑ Yes ☐ No

Applicable of Repair Welding ☑ Yes ☐ No

Retainer: None

Details



RG = 0 - 3.0 mm.
RF = 0 - 2.0 mm.
T = 1.5 – 14.22 mm.
@ = 60° ± 5 deg.

BASE METALS (QW-403)

	P-No.	1	Group no. 1	to P-no. 1	Group no. 1
OR	Specification Type and Grade	SA 106 GR.B		to	SA 106 Gr.B
OR	Chem. Analysis and Mech. Prop.	NA		to	NA
	Base Metal Thk. Range (T, mm.)	Groove	1.5 - 14.22 mm	Fillet	All thickness
	Pipe Dia. Range	Groove	All daimeter	Fillet	All thickness
	Other	None			

FILLER METALS (QW-404)

Spec. no. (SFA)	1/	5.18	2/	-
AWS no. (Class)	1/	ER 70S-G	2/	-
F-no.	1/	6	2/	-
A-no.	1/	1	2/	-
Size of Filler Metals	1/	Ø 1.6, 2.0, 2.4 mm	2/	-
Deposited Weld Metal Thk. Range (t, mm.)				
Groove	1/	14.22 mm (Max)	2/	-
Fillet	1/	All thickness	2/	-
Type-wire	1/	Solid wire	2/	-

Electrode-Flux (class): NA

Flux Trade Name: NA

Consumable Insert: None

Other: Brand name : KOBELCO TGS-50

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD

RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

59 M.8 HIGHWAY NO. 36 T. TUBMA A. MUANG RAYONG 21000

Tel. : (038) 663476 FAX: (038) 663477 Mobile : (081) 8613795

WPS no. RNC-WS-002 Rev. 0

POSITIONS (QW-405)

Position of Groove 1/ All 2/ -

Welding Progression : Up Hill Down -

Position(s) of Fillet 1/ All 2/ -

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range (NO PWHT) °C

Time Range - hr.

Other None

PREHEAT (QW-406)

Preheat Temp. Min. 10 °C

Interpass Temp. Max. 250 °C

Preheat Maintenance Cool in still air

Other None

GAS (QW-408)

	Percent Composition		
	Gas(es)	Mixture	Flow Rate
Shielding	Argon	Welding Grade	10-20 L/min
Trailing	NA	-	-
Backing	NA	-	-

ELECTRICAL CHARACTERISTICS (QW-409)

Current AC or DC 1/ DC 2/ - Polarity 1/ EN 2/ -

Amps (Range) See table below Volts (Range) See table below

Tungsten Electrode Size and Type Ø 2.4 mm / 2%Thoriated

Mode of Metal Transfer for GMAW/FCAW NA

Electrode Wire Feed Speed Range NA

Heat input (Range) NA

TECHNIQUE (QW-410)

String or Weave Bead 1/ String or Weave 2/ -

Orifice or Gas Cup Size 1/ No. 4 - 9 2/ -

Initial and Interpass Cleaning Power brushing and / or Grinding

Method of Back Gouging NA

Oscillation None

Contact Tube to Work Distance 1/ NA 2/ -

Multiple or Single Pass (per side) 1/ Multiple 2/ -

Multiple or Single Electrodes 1/ Single 2/ -

Travel Speed (Range) See table below

Peening No

Other None

Weld Layer(s)	Process(es)	Filler Metal		Current		Volt Range	Travel Speed	Heat Input
		Class	Dia.	Type Polar.	Amp.Range		Range: cm/minute	Range : kJ/mm
1	GTAW	ER 70S-G	1.6,2.0,2.4 mm	DCEN	110 - 150A	16 - 22 V	4 - 11	-
2	GTAW	ER 70S-G	1.6,2.0,2.4 mm	DCEN	130 - 170A	18 - 25 V	4 - 11	-
3 and Over	GTAW	ER 70S-G	1.6,2.0,2.4 mm	DCEN	120 - 170A	16 - 25 V	4 - 11	-
				***********************				

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD

RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

59 M.8 HIGHWAY NO. 36 T. TUBMA A. MUANG RAYONG 21000

Tel. : (038) 663476 FAX: (038) 663477 Mobile : (081) 8613795

PROCEDURE QUALIFICATION RECORD (PQR)

Company Name: RENEWCOILS ENGINEERING &SUPPLY CO.,LTD.

Procedure Qualification Record no.: RNC-PQ-002 Date: MAY 25, 2009

WPS no.: RNC-WS-002 Revision. no.: 0

Welding Process(es) 1/ GTAW 2/ -

Types (Manual, Automatic, Semi-Automatic) 1/ Manual 2/ -

Impact Test Requirement ☐ Yes ☑ No

JOINTS (QW-402)

SEE PAGE 3 OF 3

BASE METALS (QW-403)

Material Spec. SA106 to SA106
Type of Grade GR.B to GR.B
P-no. 1 Group 1 to P-no. 1 Group 1
Thickness of Test Coupon, T 7.11 mm.
Diameter of Test Coupon NPS 6"SCH.40
Other None

POSTWELD HEAT TREATMENT (QW-407)

Temperature (NO PWHT) °C
Time - hr.
Other None

GAS (QW-408)

	Gas(es)	Percent Composition Mixture	Flow Rate
Shielding	Argon	99.90%	12 L/min.
Trailing	NA	-	-
Backing	NA	-	-

FILLER METALS (QW-404)

SFA Specification 1/ 5.18 2/ -
AWS Classification 1/ ER 70S-G 2/ -
Filler Metal F.no. 1/ 6 2/ -
Weld Metal Analysis A. No. 1/ 1 2/ -
Size of Filler Metal 1/ Ø 2.4 mm. 2/ -
Other Solid wire filler metal
Deposited Weld Metal thickness, t (mm.) 1/ 7.11 mm. 2/ -

ELECTRICAL CHARACTERISTICS (QW-409)

Current 1/ DC 2/ -
Polarity 1/ EN 2/ -
Amps. 1/ See sheet 3 2/ -
Volts 1/ See sheet 3 2/ -
Tungsten Electrode Size Ø 2.4 mm/2%Thoriated
Heat input (Range) NA
Other None

POSITIONS (QW-405)

Position of Groove 1/ 6G 2/ -
Welding Progression (Uphill, Downhill) 1/ Uphill 2/ -
Other None

TECHNIQUE (QW-410)

Travel Speed 1/ See sheet 3 2/ -
String or Weave Bead 1/ Weave 2/ -
Oscillation None
Multipass or Single Pass (per side) Multipass
Single or Multiple Electrodes Single
Other None

PREHEAT (QW-406)

Preheat Temp. Ambient (~30) °C
Interpass Temp. 150 (Max.) °C
Other Cool in still air

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.
59 M.8 HIGHWAY NO. 36 T. TUBMA A. MUANG RAYONG 21000
Tel. : (038) 663476 FAX: (038) 663477 Mobile : (081) 8613795

PQR Back

PQR no. RNC-PQ-002

TENSILE TEST (QW-150)

Specimen no.	Width (mm.)	Thickness (mm.)	Area (mm^2)	Ultimate Tensile Load (N)	Ultimate Unit Stress (N/mm^2)	Character of Failure and Location
TS 174 / 2009	19.34	6.58	127.26	66,100	519.41	Failure out of weld
TS 175 / 2009	19.38	6.52	126.36	66,000	522.32	Failure out of weld

GUIDED BEND TEST (QW-160)

Specimen no.	Type & Figure No. of Bend Tests	Results	Remarks
FB 693 / 2009	Transverse Face Bend (QW 462.3 (a))	Acceptable	Open less than 0.5 mm. at weld
RB 694 / 2009	Transverse Root Bend (QW 462.3 (a))	Acceptable	No Discontinuity
FB 695 / 2009	Transverse Face Bend (QW 462.3 (a))	Acceptable	No Discontinuity
RB 696 / 2009	Transverse Root Bend (QW 462.3 (a))	Acceptable	No Discontinuity

TOUGHNESS TEST (QW-170)

Specimen no.	Notch Location	Notch Type	Test Temp. °C	Impact Value			Drop Weight Break	
				J	%Shear	Mils	Break	No Break
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-

FILLET-WELD TEST (QW-180)

Results - Satisfactory : ☐ Yes ☐ No Penetration into Parent Metal : ☐ Yes ☐

Macro - Results : NA

OTHER TESTS

Type of Test RADIOGRAPHIC TEST AS PER REPORT NO.R09-2239-2-P42

Deposit Analysis NA

Other None

Welder's name Mr. Pancha Sonsakha Clock no. - Stamp no. WT - 009

Test conducted by SIWA TESTING INSPECTION & CONSULTING CO., LTD. Laboratory Test no. R09-2239-8,9-P42

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Code.

Manufacturer RENEWCOILS ENGINEERING &SUPPLY CO.,LTD.

Date Certify

บริษัท สิวะ เทสติ้ง อินสเพ็คชั่น แอนด์ คอนซัลติ้ง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

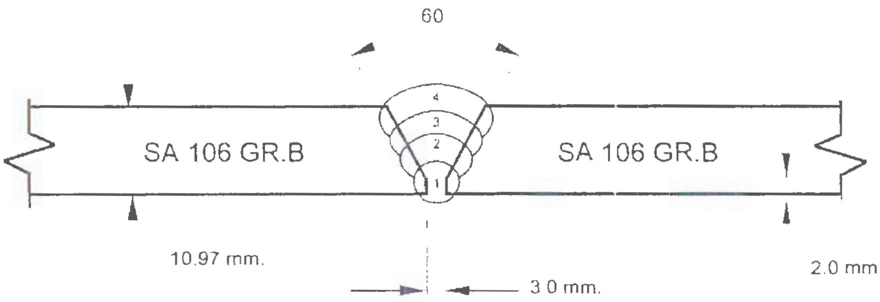
ORIGINAL

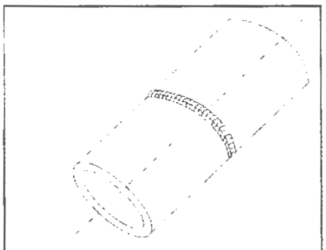
Actual Records of Welding Variable

ACC. TO ASME SECTION IX WELDING VARIABLE

WPS NO RNC-WS-002 Revision No. 0
WELDER NO. WT-009

Front side




6G Position

Type of material ☑ Pipe (NPS 6" SCH40) ☐ Plate
Backing ☐ Yes ☑ No
Retainers ☐ Yes ☑ No

PASS SEQUENCE

Side	Pass	Process	Electrode or Filler Type	Electrode or Filler Size	Amps./ Polarity	Volts	Travel Speed (cm./minute)	Heat Input (kJ/mm)	Remarks (gas flow rate)
Front	1	GTAW	ER 70S-G	Ø 2.4 mm.	125 - 133 / DCEN	18 - 20 V.	6.3	-	Ar: 12 L/Min.
	2	GTAW	ER 70S-G	Ø 2.4 mm.	151 - 154 / DCEN	20 - 22 V.	8.5	-	Ar: 12 L/Min.
	3	GTAW	ER 70S-G	Ø 2.4 mm.	152 - 154 / DCEN	20 - 22 V.	8.2	-	Ar: 12 L/Min.
	4	GTAW	ER 70S-G	Ø 2.4 mm.	138 - 140 / DCEN	18 - 20 V.	9.8	-	Ar: 12 L/Min.
				****************************					

Witnessed by : (Mr. Witsanu K.) Date : MAY 16, 2009

SUMMARY OF WELDER/WELDING OPERATOR PERFORMANCE QUALIFICATION

Manufacturer : RENEW COILS ENGINEERING & SUPPLY CO., LTD.

Project : PQR

Applicable Code : ASME SEC. IX Edition : 2007 Addenda : -

Witness Date : MAY 16, 2009 Place : RAYONG

Material Specification : SA 106 Gr.B

Electrode/Filler Metal Classification : ER 70S-G (KOBELCO TGS-50)

Welder No.	Name of Welder Welding Operator	Process	Type	Test Coupon	Test Position	Visual	RT	Tensile test	Bend test	Results
./T-009	Mr. Pancha Sonsakha	GTAW	Manual	NPS 6" SCH40	6G	ACC	ACC	ACC	ACC	PASSED
			**************************************							
	PQR no. RNC-PQ-002									
	C. [illegible] May 28, 2009 STK									

Legend : RT = Radiographic Test, NA = not applicable, ACC = Accept, REJ = Reject

Note. For complete qualification range, see attached certificate(s).

SUMMARY OF WELDER/WELDING OPERATOR PERFORMANCE QUALIFICATION

Manufacturer : RENEW COILS ENGINEERING & SUPPLY CO., LTD.

Project : WPQ

Applicable Code : ASME SEC. IX Edition : 2007 Addenda : -

Witness Date : MAY 16, 2009 Place : RAYONG

Material Specification : SA 106 Gr.B

Electrode/Filler Metal Classification : ER 70S-G (KOBELCO TGS-50)

Welder No.	Name of Welder Welding Operator	Process	Type	Test Coupon	Test Position	Visual	RT	Tensile test	Bend test	Results
WT-001	Mr. Chuchat Mongkhonsathian	GTAW	Manual	NPS 2" SCH80	6G	ACC	ACC	NA	NA	PASSED
WT-009	Mr. Pancha Sonsakha	GTAW	Manual	NPS 2" SCH80	6G	ACC	ACC	NA	NA	PASSED
		**								

C. [illegible]
May 26, 20[illegible]

Legend : RT = Radiographic Test, NA = not applicable, ACC = Accept, REJ = Reject

Note. For complete qualification range, see attached certificate(s).

FORM : WQT-SUM2 R.1, 16 Mar. 2009

SUMMARY OF WELDER/WELDING OPERATOR PERFORMANCE QUALIFICATION

Manufacturer : RENEW COILS ENGINEERING & SUPPLY CO., LTD.

Project : WPQ

Applicable Code : ASME SEC. IX Edition : 2007 Addenda : -

Witness Date : MAY 16, 2009 Place : RAYONG

Material Specification : SA 106 Gr.B

Electrode/Filler Metal Classification : ER 70S-G (KOBELCO TGS-50)

Welder No.	Name of Welder Welding Operator	Process	Type	Test Coupon	Test Position	Visual	RT	Tensile test	Bend test	Results
WT-001	Mr. Chuchat Mongkhonsathian	GTAW	Manual	NPS 6" SCH40	6G	ACC	ACC	NA	NA	PASSED
		**								

May 26,

Legend : RT = Radiographic Test, NA = not applicable, ACC = Accept, REJ = Reject

Note. For complete qualification range, see attached certificate(s).

บริษัท ศิวะ เทสติ้ง อินสเพ็คชั่น แอนด์ คอนซัลติ้ง จำกัด

SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

ORIGINAL

RADIOGRAPHIC TEST REPORT

Report No. :	R09-2239-4-P42
Test Date :	May 18, 2009
Client :	Renew Coils Engineering & Supply Co., Ltd.
Description of Work :	WPQ
Place of Inspection :	STIC
Specification :	ASME SEC. IX
Wall Thickness :	Ø 2" X 5.54 mm., GTAW, C/S, 6G
Source :	Gamma Ray
Technique :	Double Wall - Double Image
Film Type :	Kodak MX125
Total Films :	3.5" x 8.5" = 4 Film(s), 3.5" x 17" = - Film(s)

No.	Section	Weld Id.	Interpretation	Result
1	WT-001	A	Minor Porosity	Accept
2		B	No Visual Defect	Accept
3	WT-009	A	No Visual Defect	Accept
4		B	No Visual Defect	Accept
		************************************		

Operated by : Ekasit Khamta ☐ **Attached Report = Page(s)**

Reviewed/Supervised by : (Udom Panpradab/RT Level II) **Date :** MAY 21, 2009

Certified by : (Somchai Ruangsuriyakij/Project Engineer) **Date :** 21/5/2009

FORM : RT-STD1 PG1 R.2 ISSUE NO.1

บริษัท ศิวะ เทสติ้ง อินสเพ็คชั่น แอนด์ คอนซัลติ้ง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO., LTD.

TENSILE TEST REPORT

Report no. : R09-2239-8-P42 Page 1 of 1

Test Date : May 22, 2009

Client : Renew Coils Engineering & Supply Co., Ltd.	Tested Product : PQR No. RNC-PQ-002 (Welded Pipe Ø 6")	Material : SA106 Gr.B	
Project : -	Applicable Standard : ASME Sec. IX : 2007	Thickness : 7.11	mm

No.	Specimen No.	Size of Specimen (mm)	Gauge Length (mm)	Elongation (mm)	Elongation (%)	Tested Area (mm^2)	Load Yield (N)	Load Ultimate (N)	Stress Yield (N/mm^2)	Stress Ultimate (N/mm^2)	Remark
1	TS 174 / 2009	19.34 x 6.58	-	-	-	127.26	-	66,100	-	519.41	FOOW - Accept
2	TS 175 / 2009	19.38 x 6.52	-	-	-	126.36	-	66,000	-	522.32	FOOW - Accept

****

Remark : FAW - Failure at weld
FOOW - Failure out of weld

Total	2 Specimen(s)	☐ Attached Report = Page(s)	
Operated by	Nopporn Karnjanajutaphan	Reviewed and Supervised by :	(Patoomporn Phongchotip)
Certified by	(Patoomporn Phongchotip / Project Engineer)	Date : 25 MAY 2009	

FORM . Tensile Test R 13Jul07 ISSUE NO. 1

ORIGINAL

สำนักงานกรุงเทพ 195 ซอยเพชรเกษม 65 ถนนเพชรเกษม แขวง/เขตบางแค กรุงเทพฯ 10160
Bangkok Office 195 Soi Petchkasem 65, Petchkasem Road, Bangkhae, Bangkok 10160, Thailand
Tel. (66)(0) 2444-3645 (15 lines) Fax. (66)(0) 2444-3026, 2444-3027 Email. info@siwatesting.com

สำนักงานระยอง 267/306-307 ถนนสุขุมวิท ต.มาบตาพุด อ.เมือง จ.ระยอง 21150
Rayong Office 267/306-307 Sukhumvit Road, Map Ta Phut, Amphur Muang, Rayong 21150, Thailand
Tel. (66)(0) 3860-9620 (4 Lines) Fax. (66)(0) 3868-9394 E-mail. rayong@siwatesting.com Website. www.siwatesting.com

ORIGINAL

บริษัท ศิวะ เทสติ้ง อินสเป็คชั่น แอนด์ คอนซัลติ้ง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO., LTD.

GUIDED BEND TEST REPORT	Report no. : R09-2239-9-P42	Page 1 of 1
	Test Date : May 22, 2009	

Client :	Renew Coils Engineering & Supply Co., Ltd.
Project :	-
Tested Product :	PQR No. RNC-PQ-002 (Welded Pipe Ø 6")
Material Specification :	SA106 Gr.B — Thickness : 7.11 mm
Applicable Standard :	ASME Sec. IX : 2007 — Bending Angle : 180 Deg. — Mandrel Dia. : 28.0 mm

Specimen No.	Type of Bending	Size of Specimen	Fracture Location	Remark
FB 693 / 2009	Transverse Face Bend	38.0 x 7.11	Open less than 0.5 mm at Weld	Accept
RB 694 / 2009	Transverse Root Bend	38.0 x 7.11	No discontinuity	Accept
FB 695 / 2009	Transverse Face Bend	38.0 x 7.11	No discontinuity	Accept
RB 696 / 2009	Transverse Root Bend	38.0 x 7.11	No discontinuity	Accept
		**		

Total : 4 Specimen(s)	☐ Attached Report = Page(s)	
Operated by	Noppom Karnjanajutaphan	Reviewed and Supervised by : (Patoomporn Phongchotip)
Certified by	(Patoomporn Phongchotip / Project Engineer)	Date : 25 MAY 2009

FORM : Bend Test R.13Jul07 ISSUE NO. 1

สำนักงานกรุงเทพ 195 ซอยเพชรเกษม65 ถนนเพชรเกษม แขวง/เขตบางแค กรุงเทพฯ 10160
Bangkok Office 195 Soi Petchkasem 65, Petchkasem Road, Bangkhae
Bangkok 10160, Thailand
Tel. (66)(0) 2444-3645 (15 lines) Fax. (66)(0) 2444-3026, 2444-3027
Email. info@siwatesting.com Website. www.siwatesting.com

สำนักงานระยอง 267/306-307 ถนนสุขุมวิท ต.มาบตาพุด อ.เมือง จ.ระยอง 21150
Rayong Office 267/306-307 Sukhumvit Road, Map Ta Phut,
Amphur Muang, Rayong 21150, Thailand
Tel. (66)(0) 3860-9620 (4 Lines) Fax. (66)(0) 3868-9394
E-mail. rayong@siwatesting.com Website. www.siwatesting.com

บริษัท สิวะ เทสติ้ง อินสเพ็คชั่น แอนด์ คอนซัลติ้ง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

ORIGINAL

RADIOGRAPHIC TEST REPORT

Report No. :	R09-2239-2-P42
Test Date :	May 18, 2009
Client :	Renew Coils Engineering & Supply Co., Ltd.
Description of Work :	PQR
Place of Inspection :	STIC
Specification :	ASME SEC. IX
Wall Thickness :	Ø 6" X 7.11 mm., GTAW, C/S, 6G
Source :	Gamma Ray
Technique :	Double Wall - Single Image
Film Type :	Kodak AA
Total Films :	3.5" x 8.5" = 4 Film(s), 3.5" x 17" = - Film(s)

No.	Section	Weld Id.	Interpretation	Result
1	RNC-PQ-002	A-B	No Visual Defect	Accept
2		B-C	No Visual Defect	Accept
3		C-D	No Visual Defect	Accept
4		D-A	No Visual Defect	Accept
		************************************		

Operated by :	Ekasit Khamta	☐ **Attached Report** = **Page(s)**
Reviewed/Supervised by :	(Udom Panpradab/RT Level II)	**Date :** MAY 21, 2009
Certified by :	(Somchai Ruangsuriyakij/Project Engineer)	**Date :** 21/5/2009

FORM : RT-STD1 PG1 R.2 ISSUE NO.1

ORIGINAL

บริษัท ศิวะ เทสติ้ง อินสเพ็คชั่น แอนด์ คอนซัลติ้ง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

RADIOGRAPHIC TEST REPORT

Report No. : R09-2239-5-P42
Test Date : May 18, 2009
Client : Renew Coils Engineering & Supply Co., Ltd.
Description of Work : WPQ
Place of Inspection : STIC
Specification : ASME SEC. IX
Wall Thickness : Ø 6" X 7.11 mm., GTAW, C/S, 6G
Source : Gamma Ray
Technique : Double Wall - Single Image
Film Type : Kodak AA
Total Films : 3.5" x 8.5" = 4 Film(s), 3.5" x 17" = - Film(s)

No.	Section	Weld Id.	Interpretation	Result
1	WT-001	A-B	No Visual Defect	Accept
2		B-C	No Visual Defect	Accept
3		C-D	No Visual Defect	Accept
4		D-A	No Visual Defect	Accept
		**********************************		

Operated by : Ekasit Khamta
☐ **Attached Report = Page(s)**

Reviewed/Supervised by : (Udom Panpradab/RT Level II)
Date : MAY 21, 2009

Certified by : (Somchai Ruangsuriyakij/Project Engineer)
Date : 21/5/2009

FORM : RT-STD1 PG1 R.2
ISSUE NO.1